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Pricing Strategies

BUNDLING

Summary of Key Points:

1) Bundling is a combination of product line pricing and non-linear pricing. That is, we might sell products A, B, and A&B. (Example: Software bundle, Value meals) We might also sell one unit of A, 2 units of A, 3 units of A, etc. (Example: Season tickets, individual game tickets, half-season of tickets)
2) Basic Rules:
    a. Sum of Component Prices > Price of Bundle ($P_A + P_B > P_{AB}$)
    b. Marginal Contribution of Bundle > Marginal Cost of Making Bundle ($P_{AB} - P_B > c_A$)

    [Note: This is generally true, but not always!]

3) If we follow a product line interpretation, it is important to consider whether A and B are substitutes, complements, or independent. When independent, we expect all prices increase. When substitutes, we expect to see the bundle price much less than the sum of the component prices. When complements, we expect to see the bundle price closer to the sum of the component prices.

4) The benefits of bundling often follow from an inverse value of the products (or a negative correlation). Example: Value of Desserts and Main Course is $2 and $18 for customer 1 versus $17 and $3 for customer 2. Sell the bundle at $20.

5) Why add the components to this bundle? Grow the market (customers that don't buy the bundle but will buy an individual item). Shift demand to more profitable products (the firm may prefer to sell the item alone than the bundle). For example, if you have value of $0 for a dessert but $22 for a main course, you will buy the bundle of main course and dessert at $20. But if the cost of the dessert is positive, say $1.50, I would rather sell you a main course for $19 and no dessert. (see condition (2b) in basic rules)

6) Challenges: Measuring valuation of components and bundle can be difficult. To assist with this, it is critical to segment:

"Loyals": will buy, but trade off bundle vs. items. Reference product is within the product line (item alone or bundle)
"Fickles": will buy or not buy. Reference product is not buy (or buy at competing firm).
This sort of segmentation may facilitate design and/or interpretation of EVC, conjoint, or another measurement tool.

Are the products substitutes, complements or independent?

Independent Items

The simplest case is when A and B are independent products. In this case, the Bundle AB and the items are all substitutes. This leads us back to product line pricing with substitutes. We know that the bundle with the components will be priced higher than the bundle alone. More generally, the price of both the components (A or B) and the bundle (AB) will increase.

Example 1a (Independent Items): Be sure you understand WHY bundle price generally (not always) increase when a bundle is offered with *a la carte* items (mixed bundle) vs. a pure bundle. Let's look at an example to see why.

Customer	WTP Item 1	WTP Item 2	WTP Bundle
Customer 1:	$18	$2	$20
Customer 2:	$10	$13	$23
Customer 3:	$14	$9	$23
Bundle Alone:			$20
Mixed Bundle:	$18	Not offered Alone or price > $5	$23

If the firm uses a pure bundling strategy, the firm "gives up" $3 per unit ($23 vs. $20) on Customer 1 and Customer 2 in order to sell the bundle to Customer 1. A more efficient way to sell to Customer 1 is to simply sell Item 1 alone. The reason customer 1 has a low WTP for the Bundle is because this customer dislikes Item 2. The logical thing to do is to not sell them this item (particularly if our cost of Item 2 is less than a customer's WTP for Item 2).

When the firm sells the mixed bundle, it charges $18 for Item 1. Now, there is a loss of -$2 on customer 1 (Recall before we got $20, now we get $18). But we are now able to raise the bundle price to $23. This leads to an additional $3 from customer 2 and customer 3 (+$6 total). The net increase in profits is +$4 = -$2 (loss on customer 1) + $6 (gain on customer 2 and 3).

Substitutes

In the above example 3, we assume that the WTP for the bundle equals the WTP of the items. If A and B are substitutes, this implies that the value to consumers of the bundle AB may be less than the sum of the components. For example, once you own a 3-quart cooking pan, your willingness to pay for a 1.5-quart cooking pan decreases. Hence your willingness to pay for the combination of 3-quart and 1.5-quart pans is less than your willingness to pay for either of the products alone. The firm selling the bundle of cooking pans will need to lower the price of the bundle accordingly. We would expect to see the bundle price much lower than the sum of the component prices. Notice that may not effect our earlier example.

Example 1b (Substitutes): Consider the earlier example, but now assume that the WTP of the bundle is not the sum of the items.

Customer	**WTP Item 1**	**WTP Item 2**	**WTP Bundle**
Customer 1:	$18	$2	$18
Customer 2:	$10	$13	$20
Customer 3:	$14	$9	$20
Bundle Alone:			$18
Mixed Bundle:	$18	Not offered Alone or price > $2	$20

In this example, the pure bundle price is lower because the WTP of the Bundle is not the sum of the WTP of the Items. However, once we add the a la carte items, the bundle price increases from $18 to $20.

Complements

If A and B are complements, customers tend to buy both products. For example, customers that buy a suit also tend to buy a tie. The bundle AB is now a substitute with respect to A or B but the components are complements with respect to each other. This creates a much more complex problem. We do know that as complements, the component prices of A and B will tend to be lower versus stand-alone. Because many people already buy the bundle, this will drive the bundle price closer to the sum of the component prices to reduce the opportunity cost of the bundle.

One question to ask is what are the advantages of formally creating the bundle AB if customers tend to buy both products. For example, if most customers already buy coffee and donuts, offering a bundle of coffee and donuts may *lose* money. Why? If the price of the bundle is less than the price of the sum of the items, then offering the bundle leads to dilution. Many customer shift from buying both a full price (full margin) to buying the bundle at a lower price (lower margin).

Example 1c (Complements): Consider the earlier example, but now assume that the WTP of the bundle is not the sum of the items.

Customer	WTP Item 1	WTP Item 2	WTP Bundle
Customer 1:	$10	$0	$20
Customer 2:	$7	$8	$23
Customer 3:	$9	$6	$23
Bundle Alone:			$20
Mixed Bundle:	Not Offer	Not Offer	$20

In the above example, the products "work together." That is, on their own, the products are not that valuable. But together, the products are valuable. This is a natural setting for bundling.

The coffee and donuts example is similar, but the magnitude of "working together" is small (or zero). That is, people are probably not willing to pay a premium for both coffee and donuts. In contrast, people may be willing to pay a premium for an integrated hardware/software product than for the individual components.

Profit Maximization when Creating a Bundle

The graphs in the lecture notes highlight the type of customer that will select an individual item, a bundle of 2 items, or neither. Since profit maximization is our objective, where does this enter the picture?

Clearly in the case of the individual items, one maximizes $(p_i\text{-}c)*q_i$ where i represents item 1 or item 2, c is the variable cost, p_i is the price and q_i is the quantity. Profits can be maximized for each product independently if there is no interdependency. If the products are substitutes or complements, one would want to consider issues that we discussed in product line pricing.

In the case of the pure bundle, one maximizes $(p_b\text{-}c_b)*q_b$ where b represents a bundle of item 1 and item 2, $c_b = c_1 + c_2$ is the variable cost, p_b is the bundle price and q_b is the quantity demanding the bundle.

The most interesting case is with mixed bundling. The firm offers the individual items at prices p_1 and p_2 as well as the bundle at p_b. How does one begin to analyze profitability?

First, let's identify the incentive to offer mixed bundles versus pure bundles. In the example in the notes, there are two areas of customers who are now served by mixed bundling.

The shaded areas in the picture below (figure 1) represent customers who now buy only item 1 or item 2 and previously would not have bought the bundle. Thus, one incentive to offer the mixed bundle is to serve customers with low willingness to pay for either item 1 or item 2 but relatively high willingness to pay for the other item.

As review, note how the lines in the picture come from the consumer's self-selection process. A customer will select item 1 rather than the bundle if:

$$\text{Surplus from Item 1} > \text{Surplus from Bundle}$$
$$r_1 - p_1 > (r_1 + r_2) - p_b$$
$$p_b - p_1 > r_2$$

Similarly,

$$\text{Surplus from Item 2} > \text{Surplus from Bundle}$$
$$r_2 - p_2 > (r_1 + r_2) - p_b$$
$$p_b - p_2 > r_1$$

Finally, the consumer surplus must also be positive (or greater than the outside option):

$$\text{Surplus from Item 1} > 0$$
$$r_1 - p_1 > 0$$

$$\text{Surplus from Item 2} > 0$$
$$r_2 - p_2 > 0$$

These inequalities are plotted in Figure 1.

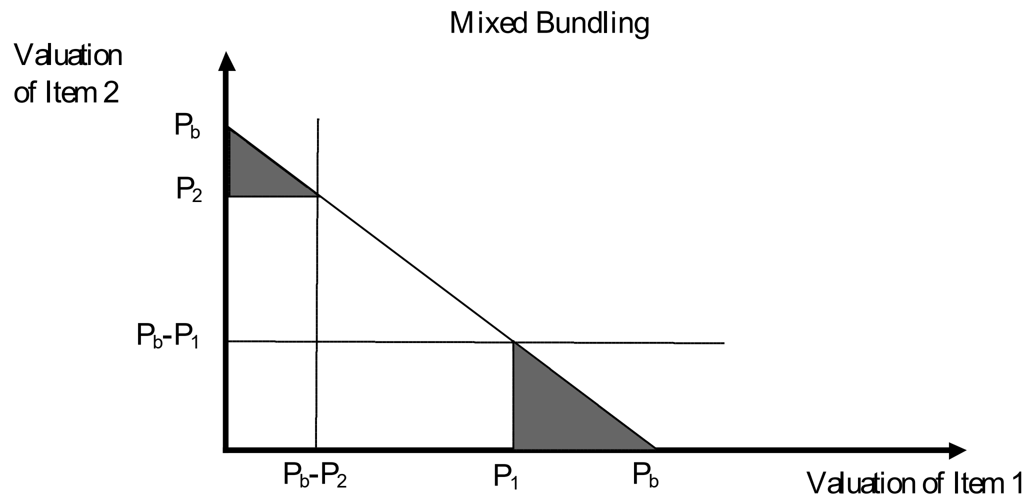

Figure 1

In the above picture, we take prices as given and focus on which customers buy which products. The issue we want to focus on now is how marginal cost influences the prices set by the firm as well as firm profits.

Suppose that the marginal costs are c_1 and c_2. The firm would like customers to select the most profitable item (item 1, item 2, or the bundle). From the firms perspective, the bundle is more profitable than item 1 if:

$$\text{Profits from Item 1} < \text{Profits from Bundle}$$
$$p_1 - c_1 < p_b - (c_1 + c_2)$$
$$p_1 < p_b - c_2$$
$$\text{or}$$
$$c_2 < p_b - p_1$$

Similarly,

$$\text{Profits from Item 2} < \text{Profits from Bundle}$$
$$p_2 - c_2 < p_b - (c_1 + c_2)$$
$$p_2 < p_b - c_1$$
$$\text{or}$$
$$c_1 < p_b - p_2$$

The above inequalities are quite intuitive. Start by thinking about selling just item 1 at price p_1. Now consider the bundle of item 1 and item 2. To create a bundle from item 1, I need to add item 2. The incremental cost of "making the bundle from item 1" is c_2. The additional revenue I earn when I sell the bundle rather than item 1 is (p_b - p_1). The inequality: $c_2 < p_b - p_1$ states that the incremental revenue earned by selling the bundle rather than the individual item (p_b-p_1) is more than the marginal cost of creating the bundle.

Another way of looking at this is to recognize that if the firm charges a high enough price on the individual items (p_1 is large so p_b-p_1 is small), it is more profitable to have customers select the individual items rather than the bundle. This provides us with our second incentive for mixed bundling. The firm may find it more profitable to sell the individual items than the bundle. Notice that when the marginal cost of the items is positive, the firm may strictly prefer to sell the individual items to some customers rather than the bundle. Example 2, which is next, illustrates this. To summarize, there are at least two advantages of mixed bundling relative to pure bundling.

Advantages of Mixed Bundling:

Selling the individual items in addition to the bundle has two possible advantages:

1. Attract New Customers (shaded area in figure 1)
2. Contribution on the Items Exceeds Contribution on Bundle

Example 2: Consider the example of a restaurant. Suppose the fixed price of a main course and dessert is $20 and the restaurant serves customers with the following willingness to pay:

Customer	**WTP Item 1**	**WTP Item 2**	**WTP Bundle**
Customer 1:	$16	$5	$21
Customer 2:	$12	$8	$20
Cost to firm	$13	$2	$15
Bundle Alone: (fixed price menu)			Price: $20 Contribution: $5 Each ($10 total)
A la Carte Only	Price: $16 Contribution: $3 ($3 total)	Price: $5 Contribution: $3 ($6 total)	
Mixed Bundle	Price: > $16	Price: $8	Bundle: $21

(Entrees + à la carte)	Contribution: $6 ($6 total)	Contribution: $6 ($6 total)

($16,$5) and ($12,$8) for (Main Course, Dessert). Suppose that main courses cost $13 to produce and desserts cost $2. If the restaurant sells the bundle at $20, both customers will buy the bundle. The total contribution is ($20-$15)*2 = $10. Notice that the best the firm could have done by selling just the individual items is $16-$13=$3 for main course and ($5-$2)*2 = $6 for dessert for a total of $9.

What if the restaurant offered individual items and the bundle? Suppose the restaurant sells the bundle at $21 and the individual items at $17 and $8. First, check the consumer's selection of items. Customer 1 ($16 main,$5 dessert) will buy the bundle as this gives zero surplus, The main meal alone and the dessert alone give negative surplus. Customer 2 ($12 main,$8 dessert) will buy just the dessert as this gives zero surplus. The bundle and the main meal alone give negative surplus. Have firm profits increased? The profits to the restaurant are ($21-$15= $6) for the bundle and ($8-$2) = $6 for the dessert for a total of $12. The restaurant does better offering the mixed bundle than the bundle alone or the individual items alone. (Note that even if we had kept the bundle price at $20 and the desserts at $7.99, profits are $10.99 vs. $10).

Notice here that we did not expand the market. The improvement in profits comes from getting customers to select items that we have higher margins on. In the above example, Customer 2 has a reservation price below the marginal cost of the main meal (WTP $12, costs us $13). Thus the restaurant may prefer to sell this type of customer just a dessert even though the customer's total willingness to pay for the bundle is greater than the cost of the bundle.

Summary of Mixed Bundling

To summarize there are two effects that influence profits when offering mixed bundles. The first effect is market expansion relative to the bundle only market. This is the shaded area in the figure. The second effect is that the firm can get customers to self-select into items that are more profitable for the firm. In the example, it is more profitable for the firm to sell only desserts to some customers and bundles to others.

Finally, remember that while we have not explicitly shown this, it should be obvious that a necessary condition for mixed bundling is that $p_1+p_2 > p_b$. If not, no one will buy the bundle.

Dilution and Stimulation in Mixed Bundling

Similar to product pricing, the firm needs to trade off stimulation and dilution in mixed bundling. To see this, consider the trade-off that occurs as the firm increases p_2. First, fewer customers purchase item 2 (the shaded area in Figure 2a). Second, customers who were buying just item 2 may start to buy the bundle. When we sold a single product, the basic trade-off was margin (p-c) versus quantity sold. In mixed bundling, we see effects similar to product-line pricing. As p_2 increases, we get the standard effect of higher margin and lower quantity sold (this is the effect of the shaded area). In the American Airlines case, this would be called "stimulation." The second effect is that customers switch from the item to the bundle. This is something like the "dilution" effect in AMR.

To sort all of this out, let's look at figures 2a and 2b. Consider just the effect of raising the price of item 2 when there is no bundle. We face the standard trade-off of higher margin and lower quantity sold. This is figure 2a. Now, let's add the bundle and analyze the effect of a price increase. In figure 2b, the dark shaded area represents lost customers who no longer purchase either item 2 or the bundle. The light shaded area represents customers who now purchase the bundle rather than just item 2.

Unlike AMR, this dilution effect is not necessarily negative. The relevant condition for item 2 will be p_b-$p_2 > c_1$ (and similarly for item 1). If the marginal cost of item 1 is relatively small, the bundle is more profitable than item 2 alone. Suppose that p_b-$p_2 > c_1$ (e.g. start with p_2 relatively low). Raising the price of item 2 will cause more customers to switch to the bundle, which is more profitable. Thus shifting customers from Item 1 only to the bundle has a positive impact on profits.

As p_2 increases further, the condition p_b-$p_2 < c_1$ may now hold. In this case a further increase in price leads to "dilution." When this condition holds, raising the price further trades off three effects.

1) Increased margin (p_2-c_2)
2) Fewer Item 2 sales
3) Dilution: Customers that Shift from Item 2 (high margin) to lower margin bundle.

On every customer that shifts, the firm gets (p_b-c_1-c_2) rather than (p_2-c_2). The Dilution effect is ((p_b-p_2) - c_1 <0).

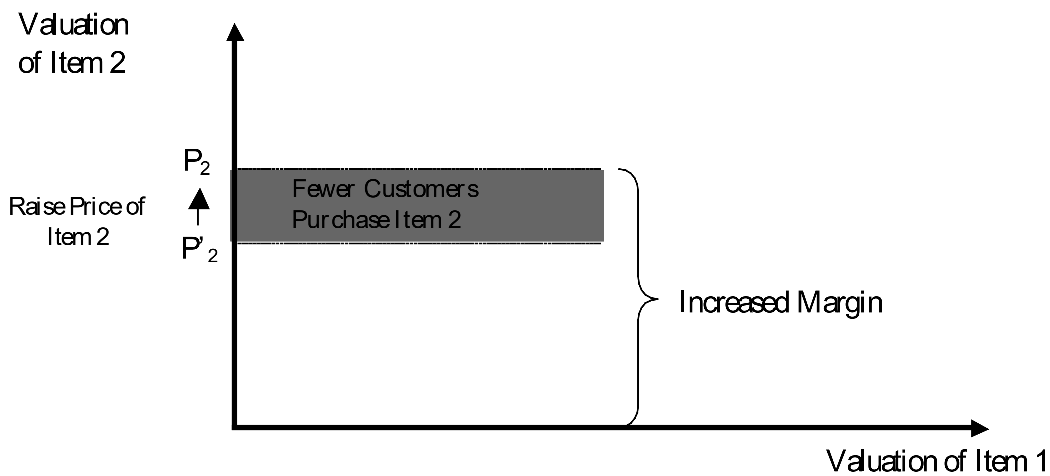

Figure 2a: Item 2 Only
Raise Price of Item 2

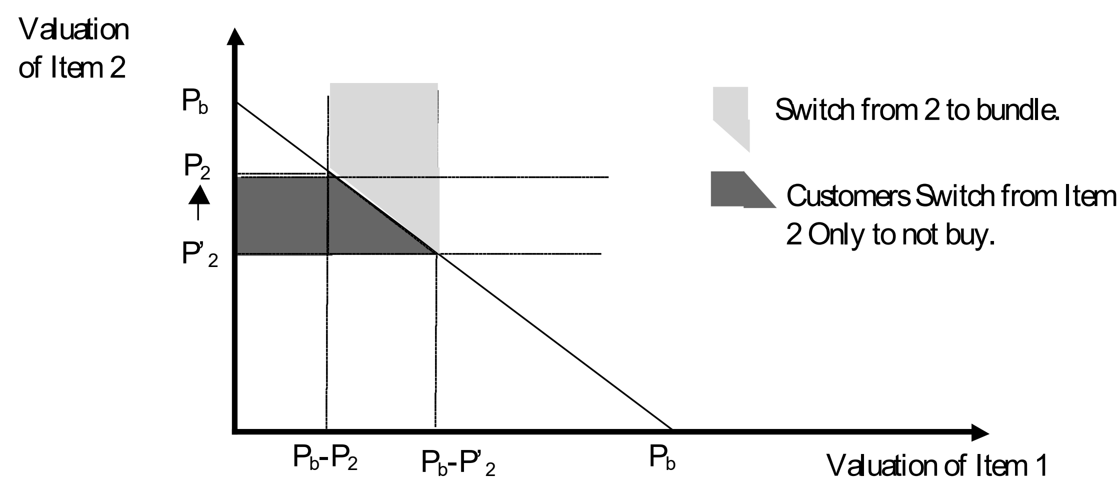

Figure 2b: Item 2 and Bundle
Raise Price of Item 2

Measurement

One of the difficulties with setting prices is that there are so many options: items alone, pure bundle, mixed bundle. With an unlimited budget, one might be able to understand how every customer trades off each option. With a limited budget, a better understanding of customer segmentation will narrow the measurement problem.

For example, Steamboat Springs sells to two types of customers: Front Range customers and Destination customers. Front Range customers drive from the surrounding area. In contrast, Destination skiers fly to Steamboat from around the United States. Market research reveals that among Front Range skiers there is an 85% intent to return. However, among Destination skiers the intent to return is only 55%.

This segmentation tells us that the Front Range skiers are much more loyal to Steamboat. Many of these customers are already committed to Steamboat. In contrast, Destination skiers trade off Steamboat vs. other ski vacations (Vail, Telluride, etc.) and other vacations (Disneyworld, Bahamas).

Suppose one was trying to modify the price of a vacation ski packages (hotel and skiing package). Among the Front Range skiers, the main concern is "Which package do I buy?" In contrast, among the Destination skiers, the main concern is "Do I visit Steamboat on my vacation?"

By understanding the next best option (reference product), we can design more effective conjoint, EVC, surveys, or market research studies. In addition, this may give us a deeper understanding of each customers willingness to pay. For example, suppose we ask customers a few questions that helps us identify what segment they are in:

Question 1: Do you live within a 4 hour drive of Steamboat?

Question 2: Have you ever visited Steamboat?

Answering "yes" to both questions tells us the customer is likely to be in the Front Range segment. In a phone survey, we might then ask a series of questions designed for Front Range skiers. These would be based on the objective of understanding Front Range skiers WTP for a longer visit or their WTP for additional days of skiing during their stay. In contrast, if customers answer "no" to either question, then we want to understand what vacations they consider and what factors might influence their decision to visit Steamboat.

A better understanding of customer segmentation narrows the measurement procedure. For example, if we use EVC we would want to understand Front Range skiers WTP for an additional day of skiing. For Destination skiers, we would want to understand their WTP for a Steamboat vacation versus another vacation. Understanding how the reference product systematically varies by segment simplifies our analysis.